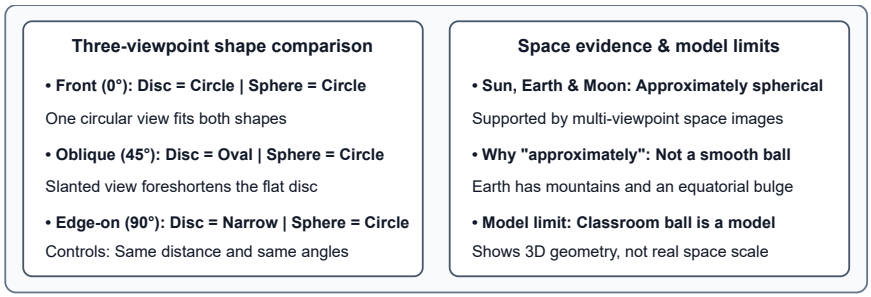
Reference sheets

Reference A · Three-viewpoint comparison: Flat disc vs Solid sphere

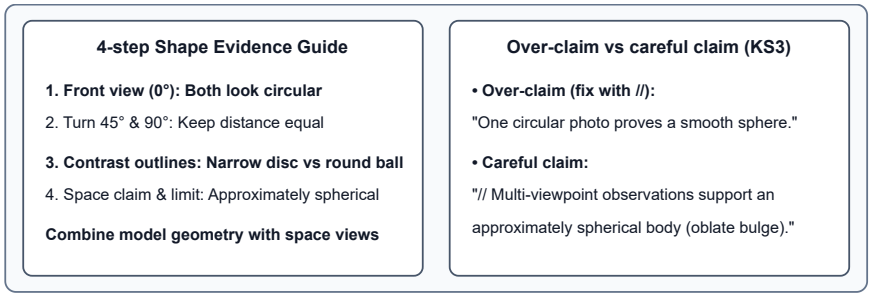


Three-viewpoint shape comparison. It compares a flat disc and a sphere at 0°, 45°, and 90°.

Reference only. Do not write on this sheet. Both objects are held at the same viewing distance across all three viewpoints.

Viewpoint angle.	Flat card disc outline.	Solid sphere (ball) outline.	What this viewpoint tells us.
1. Front view (0°).	Circular outline.	Circular outline.	Both shapes look circular from the front, so one view cannot separate them.
2. Oblique view (45°).	Oval (elliptical) outline.	Circular outline.	A slanted view foreshortens the flat disc while the sphere stays round.
3. Edge-on view (90° quarter-turn).	Narrow thin edge.	Circular outline.	Turning 90° separates the flat disc (narrow) from the 3D sphere (circular).

Reference B · Space evidence, careful claims, and model limits



Four-step Shape Evidence guide. It links 0°, 45°, and 90° views to spacecraft evidence and limits.

Evidence summary.	Key Stage 3 science meaning.	Honest model limit or careful wording.
E1 · Earth from spacecraft.	Spacecraft observations from multiple viewpoints show Earth	Earth has surface mountains and a small equatorial bulge,

Evidence summary.	Key Stage 3 science meaning.	Honest model limit or careful wording.
	as a three-dimensional round globe.	so it is approximately spherical.
E2 · Moon and Sun.	Multi-angle astronomical observations show the Moon and Sun have circular outlines from every viewpoint.	Described as approximately spherical three-dimensional bodies rather than flat discs.
E3 · Classroom ball model.	Predicts how a 3D sphere keeps a circular outline at 0°, 45°, and 90°.	Limit: the classroom ball is a geometric model, not real planetary proof or to scale.

Made by Matt

Tier pack · Independent

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use Box 1 of the arrival drawing. Distinguish a three-dimensional sphere from a two-dimensional flat disc.

Help. Compare depth in Box 1.

2. Look at Box 1. Does one circular front view at 0° prove an object is a sphere?

Help. Check what both objects look like at 0°.

3. Look at Box 2. How does a flat disc outline change at 45° and 90°?

Help. Compare the 45° and 90° outlines in Box 2.

4. Look at Box 3. Why do scientists call Earth 'approximately spherical' rather than a perfect smooth sphere?

Help. Check Box 3 for surface features and flattening.

You do tasks

1. Use Reference A. Record the outlines of a flat disc and a sphere across three controlled viewpoints (0°, 45°, and 90°). State the two fairness controls. **Staff I ☐ S ☐ C ☐**

Help. Include 0°, 45°, and 90° plus equal distance and angle.

2. Use Reference A. Why does one circular front photo fail to prove a space body is a sphere? **Staff I ☐ S ☐ C ☐**

Help. Compare the 0° row for both objects on Reference A.

3. Use Reference B. Link multi-viewpoint space evidence and one model limit. Explain why Earth is called approximately spherical. **Staff I ☐ S ☐ C ☐**

Help. Check Reference B for spacecraft views, equatorial bulge, and model limits.

Optional participation-loop prompt. Share your viewpoint comparison and limit with an adult. Hear one feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Describe one observation across viewpoints (0°, 45°, 90°) that separates a flat disc from a sphere.

2. Why must viewing distance and angle stay equal when comparing the disc and sphere?

3. Why are the Sun, Earth, and Moon called approximately spherical? State one model limit.

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Independent

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt